

Anonymous submission

Abstract

Introduction

- Mapping AI Discourse on YouTube: Cross-National Differences in Content Production, Thematic Narratives, and Audience Engagement in Brazil and the United States
- Who Talks About AI on YouTube? A Cross-National Analysis of Content Ecosystems, Narratives, and Audience Engagement
- Despite growing research on public perceptions of AI across surveys, news media, and platforms such as X/Twitter and Reddit, we know far less about how AI discourse is structured within video-based social media ecosystems.
- This is especially important in cross-national contexts, where local media systems and cultural factors may shape how global technological developments are interpreted.

Contributions: This paper introduces a large-scale dataset of AI-related YouTube videos and comments from Brazil and the United States from 2021–2025.

We make three contributions:

- We characterize the ecosystem of actors producing AI-related content on YouTube.
- We classify the dominant narratives associated with AI discourse (e.g., unemployment, scientific progress, education, ethics).
- We examine how audiences engage with different AI narratives across national contexts.

Research Questions

RQ1: Content Narratives

What narratives dominate AI-related video content in Brazil and the United States?

RQ2: Audience Reinterpretation

How do audience discussions differ from the narratives presented in videos?

RQ3: Cross-National Divergence

How do creator narratives and audience reinterpretations differ across national contexts?

RQ4: Engagement Dynamics

Which AI narratives generate the highest levels of audience engagement?

Methodology

Our analytical pipeline consists of five stages that capture both content production (video transcripts) and audience reinterpretation (comments).

Stage 1: Data Collection

The videos were collected via the YouTube Data API v3, focusing on channels from Brazil and the United States in order to compare how artificial intelligence is represented across different sociotechnical and linguistic contexts. The initial selection prioritized journalistic channels, based on prior classifications of credibility and political bias: Locatelli et al. (2024) for Brazil, and the AllSides portal¹ for the United States. To broaden the thematic and discursive diversity of the sample, we additionally included science communication, technology, educational, and podcast channels, identified through public rankings of engagement and audience².

acho que tirar essa parte de political bias porque nao usamos isso, ver se o marcelo tem um artigo mais atual que usa esse dataset –Marisa

The final sample encompasses a diverse range of formats, including news reports, analyses, interviews, opinion pieces, and educational content—allowing us to observe both more informational approaches and enthusiastic or critical positions. This diversity is essential for capturing different forms of narrative construction about AI on the platform.

Data collection spans from January 1, 2021, to XX, 2025, covering both the pre- and post-launch phases of ChatGPT (November 2022), often cited as a milestone in the popularization of generative AI.

Videos were collected from channels across multiple categories, including news media, science communication, tech-

¹<https://www.allsides.com/media-bias/media-bias-chart>

²<https://socialblade.com/youtube/top/category/tech/mostsubscribed> and <https://www.tecmundo.com.br/internet/242330-10-canais-youtube-curte-tecnologia.htm>

nology creators, educational channels, podcasts and interviews, as well as institutional talks such as TED/TEDx and university events. For each video, we collected the automatically generated transcript via the API, along with metadata including upload date, title, description, view count, like count, comment count, and channel-level information. For audience analysis, we collected all available comments and replies.

Filtering

All videos were screened for thematic relevance using an automated classification procedure. Each video was assessed based on its title and description using a prompt submitted to *XXX model –Marisa*. The prompt instructed the model to determine whether the video’s content was primarily concerned with artificial intelligence, generative AI, or closely related topics *VERIFICAR SE ESSA DESCRIÇÃO ESTÁ CORRETA –Marisa*. Videos were classified as *ESPECIFICAR CATEGORIAS: e.g., "relevant" or "not relevant"; ou categorias mais granulares –Marisa*. The full prompt is provided in Appendix *XX –Marisa*.

To assess the reliability of this automated classification, a stratified random sample of *KK –Marisa* videos was manually reviewed by *YY –Marisa* annotators, who independently assigned relevance labels following the same criteria. Inter-annotator agreement was measured using *MÉTRICA: e.g., Cohen’s kappa / percentage agreement –Marisa*, yielding a score of *UU –Marisa*, indicating *INTERPRETAR: e.g., substantial / moderate agreement –Marisa*. This procedure excluded *NN –Marisa* videos (*PERCENTUALJ –Marisa*%) from the English corpus and *UU –Marisa* videos (*PERCENTUALJ –Marisa*%) from the Portuguese corpus, resulting in a pre-cap total of 10,967 (EN) and 6,070 (PT) videos.

To manage the considerable variation in channel productivity, where a few channels accounted for a disproportionate share of the total video corpus, we applied a per-channel cap of 200 videos. For channels exceeding this threshold, we retained the 200 videos with the highest comment counts. This selection criterion is theoretically grounded in our analytical focus: since audience reactions and discursive impressions constitute the primary object of analysis, prioritizing videos that generated the greatest volume of public commentary maximizes the availability of interpretable data per video. The cap was applied separately within each language corpus (EN and PT) to preserve the structural integrity of each ecosystem while ensuring that no single channel could dominate the analysis. This procedure reduced the English corpus from 10,967 to 4,101 videos and the Portuguese corpus from 6,070 to 2,719 videos.

Fazer a mesma curva para o volume de comentarios –Marisa

Stage 2: Transcript Preprocessing

Each video transcript is treated as a single document.

We preprocess transcripts by:

- removing duplicates
- removing non-English/non-Portuguese noise
- filtering very short transcripts

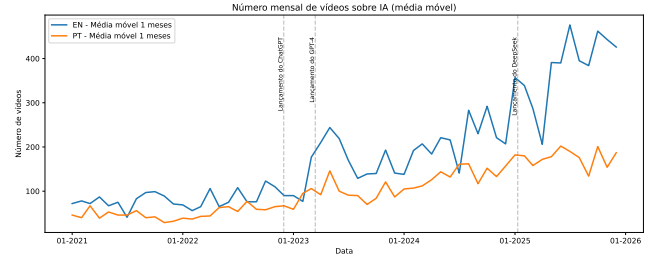


Figure 1: Monthly moving average of the number of videos about AI. Events such as the launch of ChatGPT, GPT-4, Gemini, and Sora are indicated.

Category	EN	EN Δ	PT	PT Δ
Podcast Geral	0.00 → 0.15	↑↑↑	0.22 → 0.16	↓
Institucional Geral	0.11 → 0.15	↑	nan → nan	≈
Jornalístico Tecnologia	0.26 → 0.28	≈	0.57 → 0.56	≈
Creator Tecnologia	0.01 → 0.01	≈	0.05 → 0.07	≈
Jornalístico Geral	0.62 → 0.41	↓↓↓	0.03 → 0.02	≈
Creator Geral	nan → nan	≈	0.13 → 0.18	↑

Table 1: Changes in the distribution of AI-related videos by channel category before and after the release of ChatGPT.

- standardizing punctuation
- removing automated transcript artifacts

We then generate text embeddings for each transcript to capture semantic similarity.

Stage 3: Video Narrative Identification

We identify dominant AI narratives discussed in video content using topic modeling and LLM-assisted labeling.

Potential narratives include:

- scientific progress
- productivity and workplace efficiency
- labor displacement
- education and learning
- creativity
- regulation and governance
- AI ethics
- existential risk / AGI concerns
- misinformation and deepfakes
- geopolitical competition
- everyday AI applications

For each video, we assign:

- dominant topic
- topic distribution
- narrative label

Video Topic Modeling

Each video transcript is treated as a document. We identify dominant AI narratives discussed in video content.

Outputs include:

- dominant topic
- topic distribution
- country
- channel category
- engagement metadata

Comment Topic Modeling

To reduce noise from short-form comments, we aggregate all comments from the same video into a single document.

This enables direct comparison between:

- video narratives
- audience narratives

Video-Comment Alignment

We measure whether audiences:

- reinforce video narratives
- shift discussions toward new concerns
- contest creator framing

We estimate alignment using topic overlap and semantic similarity measures.

Cross-National Comparison

We compare:

- topic prevalence
- audience reinterpretation
- engagement patterns

across Brazil and the United States.

Engagement Analysis

We examine whether specific narratives generate disproportionate attention using:

- views (reach)
- comments (discussion)
- comment-to-view ratio

Related Work

Public Perceptions of AI

Prior research has examined public attitudes toward AI, including trust, automation anxiety, technological optimism, and societal concerns.

AI Discourse on Social Media

Existing work has primarily focused on discourse about AI on Twitter/X, Reddit, news media, and search behavior platforms.

YouTube as a Sociotechnical Platform

YouTube differs from other platforms because it combines content production, algorithmic distribution, and audience participation.

Cross-National Technology Narratives

Prior research suggests that technological narratives are shaped by local contexts, motivating our comparison between Brazil and the United States.

Results

- Dominant video narratives differ across countries
- Audience discussions frequently diverge from creator narratives
- Social concerns are more prominent in comments than transcripts
- Brazilian and U.S. audiences reinterpret AI differently

Visualizations

Figure 1 : topic prevalence by country (videos). Talvez heatmap

Figure 2: topic prevalence by country (comments): heatmap

Figure 3: video topic → comment topic transitions (ver qual visualizacao)

Figure 4: engagement by topic: boxplot

Figure 5: temporal evolution pre/post ChatGPT: stacked area chart

Discussion

We argue that YouTube is not simply a passive reflection of public AI discourse. Instead, it actively shapes public understanding through platform incentives, creator ecosystems, and audience participation.

While AI narratives may be globally shared, their interpretation remains locally shaped.

Limitations

- YouTube users are not representative of national populations
- Channel selection bias
- Comment participation bias
- Transcript quality limitations

References

Locatelli, M.; Calais, P.; Miranda, M.; Junho, J.; Muniz, T.; Meira Jr., W.; and Almeida, V. 2024. Topic Shifts As a Proxy for Assessing Politicization in Social Media. In *ICWSM'24*.